

Multimodal Emotion Perception for VR Dialogue: Towards Improved User Experience

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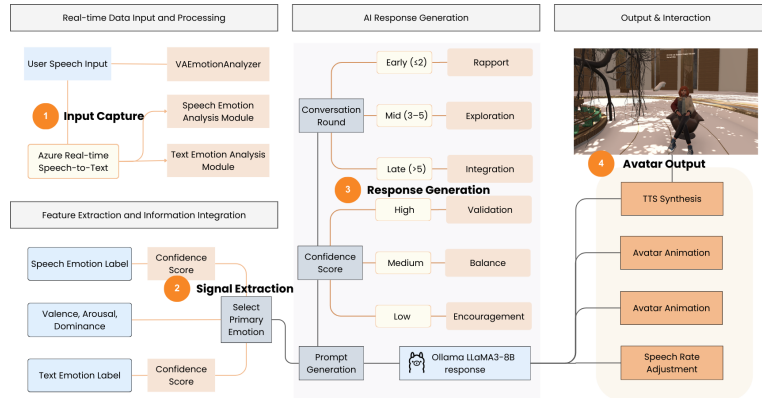


Figure 1: Workflow of the emotion-aware dialogue system.

ABSTRACT

Emotion plays a key role in user experience in virtual reality, yet the impact of affect-adaptive dialogue remains underexplored. We present a VR dialogue system that integrates speech- and text-based affect cues through a lightweight fusion method to adapt agent behavior. In a within-subjects study with twenty participants, the adaptive system led to higher user satisfaction and greater increases in positive affect compared to a non-adaptive baseline. The results suggest that affect-adaptive responses can enhance subjective experience even when emotion recognition signals are imperfect.

Index Terms: Virtual Reality, Affective Computing, Multimodal Emotion Recognition, Dialogue Systems, User Experience

1 INTRODUCTION

Understanding users' emotional states is essential for designing interactive systems in virtual reality. Prior work has shown that immersion and presence strongly shape affective experience [10, 9], yet many VR dialogue systems still rely on semantic analysis and fixed response patterns.

Recent advances in speech-based affect recognition [1] and text-based affect analysis [8] enable systems to extract multimodal emotional cues and adjust responses accordingly. However, the integration of such multimodal affective information into VR dialogue systems remains underexplored, and prior work highlights the importance of evaluating recognition and adaptation jointly rather than separately [3, 6].

This study examines how multimodal affective information shapes VR dialogue. We developed a system that combines speech- and text-derived emotion indicators [1, 8], integrates them through a simple fusion rule, and uses the resulting signal to adjust the agent's verbal and non-verbal behaviors. Adaptation was implemented through small adjustments in tone, elaboration depth, and

facial expression intensity.

The dialogue system uses a locally deployed LLaMA3 model [5] to generate adaptive responses. A within-subjects study compared this affect-adaptive system with a non-adaptive baseline matched for conversational structure.

The study examined user-reported experience, model-estimated emotional patterns, and the relation between system-generated affect labels and subjective impressions. The findings provide insight into how affect-adaptive behavior influences VR interaction and offer implications for designing more responsive conversational agents.

2 METHODS

2.1 System Architecture

The system integrates speech- and text-based affect analysis to support adaptive dialogue in virtual reality. Spoken input was transcribed and analyzed using a Wav2Vec2-based model to extract speech-derived emotion labels, while transcribed text was processed with DistilBERT to obtain text-based emotion labels [1, 8]. Both modalities produced categorical predictions from a shared label set (SAD, ANGRY, DISGUST, FEAR, HAPPY, NEUTRAL). In addition, continuous valence-arousal-dominance (VAD) estimates were derived using lexicon-based affective norms [7].

In the adaptive condition, speech-based labels, text-based labels, and VAD estimates were jointly enabled. A lightweight confidence-based fusion rule selected the dominant affective signal, which conditioned the agent's verbal and non-verbal behavior through small adjustments in response tone, elaboration depth, and facial expression intensity. Dialogue content and turn structure were generated using a locally deployed LLaMA3 model [5]. In the baseline condition, conversational structure and expressive capabilities were identical, but all affective inputs were disabled, resulting in non-adaptive responses. All system components were executed locally to minimize latency and ensure reproducibility (see Fig. 1).

2.2 Experimental Setup

The agent presented synchronized facial expressions, speech, and subtitles, and followed a fixed dialogue structure consisting of in-

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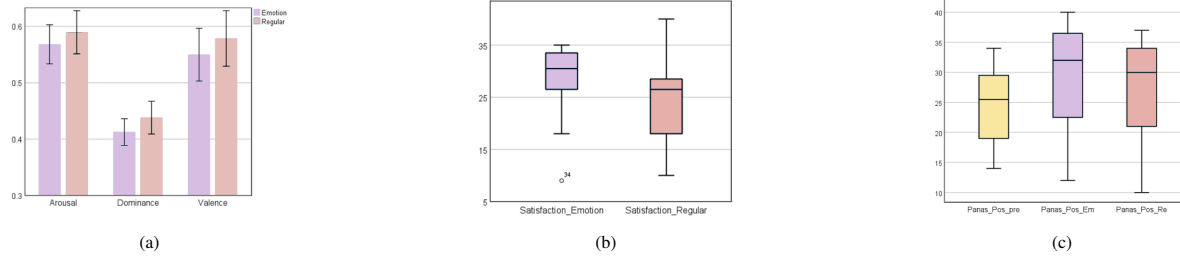


Figure 2: (a) System-generated Valence, Arousal, and Dominance. (b) User satisfaction across systems. (c) PANAS Positive Affect (pre-test vs. two systems).

roduction, reflection, and closing stages. Participants engaged in a semi-structured conversational task to elicit reflective and expressive responses, maintaining consistency across conditions [4, 12].

Twenty participants (ages 18–36) completed two five- to eight-minute conversations, one per condition, in a fully counterbalanced within-subjects design. Pre-test measures included GAD-7, PANAS, and CEQ. After each session, participants completed PANAS and satisfaction ratings, followed by a comparative preference measure. The study received IRB approval, informed consent was obtained, and all data were anonymized prior to analysis.

2.3 Measures

Self-report measures included PANAS [11], anxiety (GAD-7), expectations (CEQ [2]), satisfaction, and preference. System-derived measures included valence–arousal–dominance and emotion label distributions.

3 RESULTS

System-generated affect estimates differed modestly between conditions. The baseline condition produced slightly higher valence ($M = 0.58$ vs. 0.55 ; $t(19) = -2.89$, $p = .009$) and arousal ($M = 0.59$ vs. 0.57 ; $t(19) = -2.33$, $p = .031$), while dominance did not differ significantly ($p = .144$; see Fig. 2a). These differences were not reflected in participants’ evaluations.

In contrast, subjective measures favored the adaptive system. Participants reported higher satisfaction ($M = 28.8$ vs. 24.0 ; $t(19) = 2.59$, $p = .018$; see Fig. 2b) and higher positive affect ($M = 29.4$ vs. 24.7 , $p = .002$; see Fig. 2c), with effects in the moderate range ($d \approx 0.6$). Negative affect decreased from pre-test in both conditions, with no significant between-system difference.

Associations between system-derived and self-reported affect were moderate: valence correlated with PANAS positive affect ($r = .44$, $p = .005$), and arousal correlated with negative affect ($r = .49$, $p = .001$), whereas dominance showed no meaningful correspondence.

Baseline anxiety moderated affective change, with higher-anxiety participants showing larger reductions in negative affect ($M = -9.31$ vs. -5.75 ; $F(1,36) = 8.53$, $p = .006$). Given the exploratory nature of this study, results are reported without multiple comparison correction; non-parametric tests yielded similar trends.

4 CONCLUSION

This work examined how multimodal affect recognition influences interaction quality in VR dialogue. The study introduced a system that integrates speech- and text-based affect cues through a lightweight fusion method and compared it with a non-adaptive baseline. The system is evaluated as an interaction design probe rather than a therapeutic intervention, and no clinical claims are made.

The adaptive system yielded higher satisfaction and larger gains in positive affect, while both systems reduced negative affect.

System-derived affect estimates aligned only partially with self-reports, indicating that recognition accuracy alone cannot account for users’ subjective experience. Participants with higher baseline anxiety benefited more in terms of negative affect reduction.

The study is limited by its sample size and reliance on a single recognition pipeline. Future work may explore additional modalities and longer interaction periods.

Overall, the findings suggest that affect-adaptive behavior can enhance VR dialogue even when recognition signals are imperfect, highlighting the importance of integrating adaptation alongside affect estimation.

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